

# README

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## constitution/scripts/ multitrack – generic multi- track development engine

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Classification: universal (§            .            .            ) – project-agnostic  
engine; zero project-specific literals (no consumer  
hardware / vendor / host / package names anywhere in this  
directory, per §            .            .            (B)).

Project-agnostic runtime for the §            .            .            /  
§            .            .            multi-track development system: worktree-  
per-alias resolution, the exactly-once claim registry  
(§            .            .            (A)), the capability-aware deadlock-proof  
device-lock (§            .            .            .            .            (B)), the orchestrator bind/  
fallback/cooldown lease, the cwd-hook contract, the single-  
builder rebuild queue, and the tmux launch entrypt. Every  
consuming project inherits this directory verbatim  
(§            .            .            (B) decoupling, §            .            .            canonical-root)  
and supplies ONLY its own per-host config data.

**What lives here (GENERIC) vs what the consumer supplies (PROJECT DATA)**

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| File                                         | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>multitrack_config.sh</code>            | SOURCED library: per-host YAML loader + live-drive detect (by serial, \$ <code>MT_SERIAL</code> . <code>MT_SERIAL</code> ) + protect-guard + track-plan. Provides <code>mt_repo_root</code> / <code>mt_config_dir</code> / <code>mt_config_file</code> / <code>mt_resolve_host</code> / <code>mt_load_config</code> / <code>mt_load_pool</code> + the scalar/list accessors <code>mt_config_conductor</code> / <code>mt_config_fallback_signatures</code> . |
| <code>multitrack.sh</code>                   | Entrypoint ( <code>status</code> / <code>up</code> ).                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>multi-track_resolve_worktree.sh</code> | Alias → worktree resolution ( <code>resolve</code> ), alias → track resolution ( <code>track</code> , used by bind-on-start), and the <code>map</code> table. Honors the <code>conductor:</code> key (conductor stays on /home), orchestrator binding (most-recent wins), else positional map.                                                                                                                                                              |
| <code>multitrack_registry.sh</code>          | flock TSV get/set over <code>&lt;repo&gt;/ws_state/streams.tsv</code> .                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>multitrack_claim.sh</code>             | Exactly-once <code>item→track</code> claim + TTL reap ( \$ <code>MT_SERIAL</code> . <code>MT_SERIAL</code> (A) ).                                                                                                                                                                                                                                                                                                                                           |
| <code>multitrack_device_lock.sh</code>       | Capability-aware deadlock-proof device-lock ( \$ <code>MT_SERIAL</code> . <code>MT_SERIAL</code> (B) ).                                                                                                                                                                                                                                                                                                                                                     |
| <code>multitrack_cwd_hook.sh</code>          | Thin toolkit adapter: prints the alias's bound worktree AND fires a best-effort, non-fatal, fully-detached <code>orchestrator bind</code> (bind-on-start, \$ <code>MT_SERIAL</code> . <code>MT_SERIAL</code> ), fallback monitor, AND constitution auto-sync for the resolved track (never for the conductor).                                                                                                                                              |

| File                                           | Role                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>multi-track-constitution_sync.sh</code>  | <code>\$ . . constitution auto-sync: on every track activation, ancestor-guarded <b>fast-forward-only</b> advance of the activated worktree's <code>constitution/</code> submodule to latest <code>&lt;remote&gt;/main</code> (WIP-preserving, never <code>force/reset/rewind</code>; never <code>git submodule update</code> ). Keeps the constitution "always up to date with main, Everywhere!"</code> |
| <code>multi-track_alias_orchestrator.sh</code> | Alias <code>track bind / fallback / cooldown lease</code> .                                                                                                                                                                                                                                                                                                                                               |
| <code>multitrack_build.sh</code>               | Single-builder FIFO rebuild queue.                                                                                                                                                                                                                                                                                                                                                                        |

The consumer supplies its per-host config file (the ONLY place project strings live): `track→mount` paths, project directory name, device serials + capabilities, the `alias` account map, priority/feature list. In the reference consumer that file is `config/multitrack/<hostname>.yaml` .

## Config contract – how a generic script finds project config

A generic script self-locates BOTH roots with zero hardcoding:

- **Constitution root** – via `constitution/find-constitution.sh` (honors an explicit `CONSTITUTION_DIR` override, else walks up for `constitution/Constitution.md` / `submodules/constitution/Constitution.md` , else follows the git superproject pointer). These scripts do not need it at runtime (they are self-contained), but it is the canonical self-location primitive when a caller needs the constitution root.

- **Project/repo root** – `MT_REPO_ROOT` (explicit override) else the script's `../..`. Because these scripts live at `constitution/scripts/multitrack/`, their `../..` resolves to the constitution submodule, NOT the consuming project. A consumer therefore MUST set `MT_REPO_ROOT` (or the explicit `MT_CONFIG` full path) so config, `.ws_state`, and the tmp namespace resolve against the project, not the constitution. The reference launcher exports `MT_REPO_ROOT` for every invocation.
- **Per-host config file** – resolved by `multitrack_config.sh` as:
  - . `MT_CONFIG` – explicit full path to a config file (wins), else
  - . `$MT_CONFIG_DIR/<host>.yaml` – where `MT_CONFIG_DIR` defaults to `$(mt_repo_root)/config/multitrack` and `<host>` = `MT_HOST` else the system hostname ( `/etc/hostname` , else the first of the machine-id). An unreadable `<hostname>.yaml` falls back to `<machine-id-16>.yaml`.

Schema: `schema_version: 1` , **one file per host.**

The `MT_*` environment contract (retained  
verbatim, no renames)

| Env                | Meaning                                               | Default                                                                                       |
|--------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| MT_REPO_ROOT       | project/repo root                                     | script <code>../..</code> (⇒ set this from the constitution location)                         |
| MT_CONFIG          | explicit config-file path (overrides host resolution) | —                                                                                             |
| MT_CONFIG_DIR      | per-host config directory                             | <code>\$(mt_repo_root)/config/multitrack</code>                                               |
| MT_HOST            | host key for <code>&lt;host&gt;.yaml</code>           | system hostname → machine-id-                                                                 |
| MT_ALIAS_DIR       | orchestrator runtime dir (bindings/cool-downs/events) | <code>\${XDG_RUNTIME_DIR:-/tmp}/\$(basename "\$MT_REPO_ROOT")/multitrack/aliasorch</code>     |
| MT_LOCK_DIR        | device-lock runtime dir                               | <code>\${XDG_RUNTIME_DIR:-/tmp}/\$(basename "\$MT_REPO_ROOT")/multitrack/devicelock</code>    |
| MT_REGISTRY_LOCK   | streams-registry flock                                | <code>\${XDG_RUNTIME_DIR:-/tmp}/\$(basename "\$MT_REPO_ROOT")/multitrack/registry.lock</code> |
| MT_STREAMS_TSV     | streams registry TSV                                  | <code>\$MT_REPO_ROOT/.ws_state/streams.tsv</code>                                             |
| MT_WS_STATE        | per-repo state dir                                    | <code>\$MT_REPO_ROOT/.ws_state</code>                                                         |
| MT_WORKTREE_SUBDIR | worktree subdir under a track mount                   | <code>\$(basename "\$MT_REPO_ROOT")</code>                                                    |

| Env                                                                                   | Meaning                                                                 | Default                                                          |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------|
| MT_ALIAS_ROSTER                                                                       | alias roster<br>override<br>( name:kind,... )                           | config aliases:<br>block, else built-in                          |
| MT_BUILD_COMMAND                                                                      | build command<br>string (the<br>build.command<br>config key)            | auto-detect scripts/<br>build_maxres.sh else<br>scripts/build.sh |
| MT_BUILD_MIN_FREE_GB                                                                  | pre-build free-<br>space floor (the<br>build.min_free_gb<br>config key) | 158                                                              |
| MT_LOCK_WAIT /<br>MT_DEFAULT_TTL /<br>MT_CLAIM_TTL /<br>MT_ALIAS_TTL /<br>MT_COOLDOWN | flock/lease<br>timers                                                   | see each script header                                           |

No `MT_*` var was renamed when this engine was lifted from the reference consumer – a rename would break the live host and every paired test (`$ . .`).

## Consumer config keys (in `config/multitrack/<host>.yaml` )

Beyond `host` / `protected_drives` / `tracks` / `device_pool` / `lease_policy` / `aliases` , the config carries two OPTIONAL keys read by the engine. Both are **consumer data** (`$ . . (B)`) – the engine never hardcodes their values.



| Key                            | Shape                                                                 | Read by                                                                                          | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| conduct-<br>or:                | top-<br>level<br>scalar<br>(alias<br>name)                            | mt_con-<br>fig_con-<br>ductor →<br>the re-<br>solver                                             | \$ . . auto-con-<br>ductor. The named alias is<br>NEVER worktree-bound – its<br>session stays on the shared /<br>home checkout ( resolve<br>prints nothing / exit ,<br>and bind-on-start never binds<br>it). <b>Empty string (default) ⇒<br/>no alias is special-cased</b><br>(the /home session is simply<br>whichever session is un-<br>bound). Consumer-overridable<br>(e.g. conductor: claude3 , or a<br>provider alias to dedicate a<br>conductor account).                                           |
| fall-<br>back.sig-<br>natures: | block-<br>style<br>list<br>under a<br>top-<br>level<br>fall-<br>back: | mt_con-<br>fig_fall-<br>back_signa-<br>tures →<br>the<br>(PWU- )<br>rate-lim-<br>it monit-<br>or | \$ . . auto-fall-<br>back. The rate-limit tran-<br>script signatures the monitor<br>matches to detect a usage/<br>quota limit and trigger or-<br>chestrator fallback . Surround-<br>ing single/double quotes are<br>stripped (values that them-<br>selves contain " are single-<br>quoted in YAML). <b>Consumer-<br/>overridable</b> – a future rate-<br>limit wording change is a<br>one-line pin here, not a code<br>change. Empty/absent ⇒ no<br>signatures (the monitor de-<br>cides its own default). |

Reference-consumer seed (both keys are consumer data, so they live in the project config, never in this engine):

```
conductor: ""           # empty ⇒ /home session is conductor
fallback:
  signatures:
    - '"apiErrorStatus":429'
    - '"isApiErrorMessage":true'
```

## Runtime namespace derivation

The tmp runtime dirs ( `aliasorch` / `devicelock` / `registry.lock` ) derive their top-level namespace from `$(basename "$MT_REPO_ROOT")` (via `mt_repo_root` ), so a consumer's runtime state is namespaced by its own project directory name – no project literal is baked into the engine.

## Quick start (consumer)

```
# from the project root, with a config/multitrack/<hostname>.yaml present:
MT_REPO_ROOT="$PWD" bash constitution/scripts/multitrack/
    multitrack_resolve_worktree.sh map
# or point at an explicit config file:
MT_CONFIG=config/multitrack/<host>.yaml MT_REPO_ROOT="$PWD" \
    bash constitution/scripts/multitrack/multitrack_resolve_worktree.sh
    resolve claude1
```

## Genericization delta vs the reference consumer

Lifted from the reference project's `scripts/multitrack/` per `$ . .` with a small, mechanical delta (three executed-code fixes + comment-literal scrub):

- . **Namespace literal** – the tmp-namespaced defaults in `multitrack_registry.sh` , `multitrack_resolve_worktree.sh` , `multitrack_device_lock.sh` , and `multitrack_alias_orchestrator.sh`

derive the top-level dir from `$(basename "$MT_REPO_ROOT")` instead of a hardcoded project name (orchestrator + resolve share one runtime dir, so both are fixed in lockstep).

- . **Build command** – `multitrack_build.sh` reads the `build.command` config key via `MT_BUILD_COMMAND`, falling back to a documented auto-detect default.
- . **Item-id param** – `multitrack_claim.sh`'s user-facing `atm-id` label is the scheme-agnostic `item-id` (the positional arg accepts any stable id – no named flag existed, so no back-compat alias is needed).

Consumer-side DATA (`config/multitrack/<host>.yaml`, serials, mount paths) stays in the consuming project and is NEVER copied here.